Four Topology MOSbius

All team members are undergraduate students at Columbia University with a graduation date of May 2027.

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Team Member: Songhang Li
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Project Information

Goal: To design and fabricate a chip containing several of the most common amplifier topologies, all with accessible pins, so that a student can wire up, test, characterize, and compare each amplifier

Application: Write a small lab manual to go with chip (keep in mind during design, but can probably come after submission of design and layout). Students are able to measure DC Gain, GBW, Slew Rate, Stability, PSSR, etc. for each amplifier and learn trade-offs through measurement and experimentation

Project Information

Design:

- We will have four on-chip amplifiers with the following topologies: 5-Transistor OTA (Designed by Songhang), Two-Stage Miller OTA, maybe with compensation cap accessible via pins for stability experimentation (Designed by Max), Telescopic Cascode OTA (Designed by Manuel), and Folded Cascode OTA (Designed by Eli)
- All amplifiers will share a master bias current set by an external resistor to ground (change resistor value to alter bias current). Working mechanism: force some node on chip to a set voltage (diode-connected, etc.) and allow a resistor to be placed from this pin to force some bias current
- All amplifier inputs, outputs, and compensation nodes (when applicable) are pinned out. Students are able to wire inverting amplifiers with off-chip resistors, unity-gain buffers, etc.

Project Information

Other Design Notes:

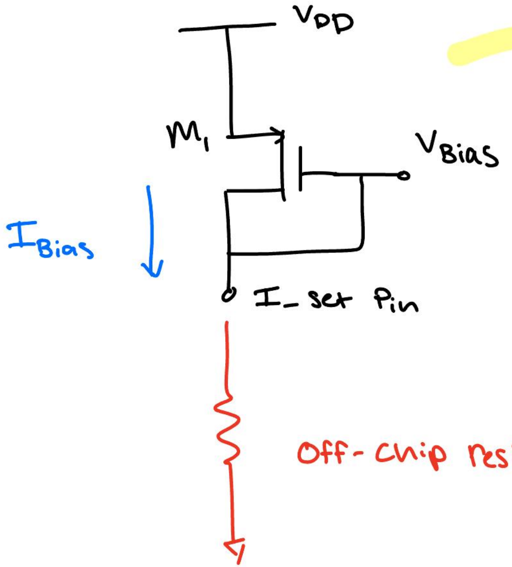
- Amplifiers will need some enable pin, so that only one is to be biased at a time
- We will also likely need some on-chip supply decoupling, also ESD protection

In thinking about what we want our chip to look like, we are considering: trying to have a block for everyone to design and layout in accordance with their experience (although we can all collaborate/help), trying to have a system that is interesting, useful, but also not too complex, or something that we cannot reasonably accomplish

Project Information

Bias Current Mechanism:

Mechanism for setting I_{Bias}



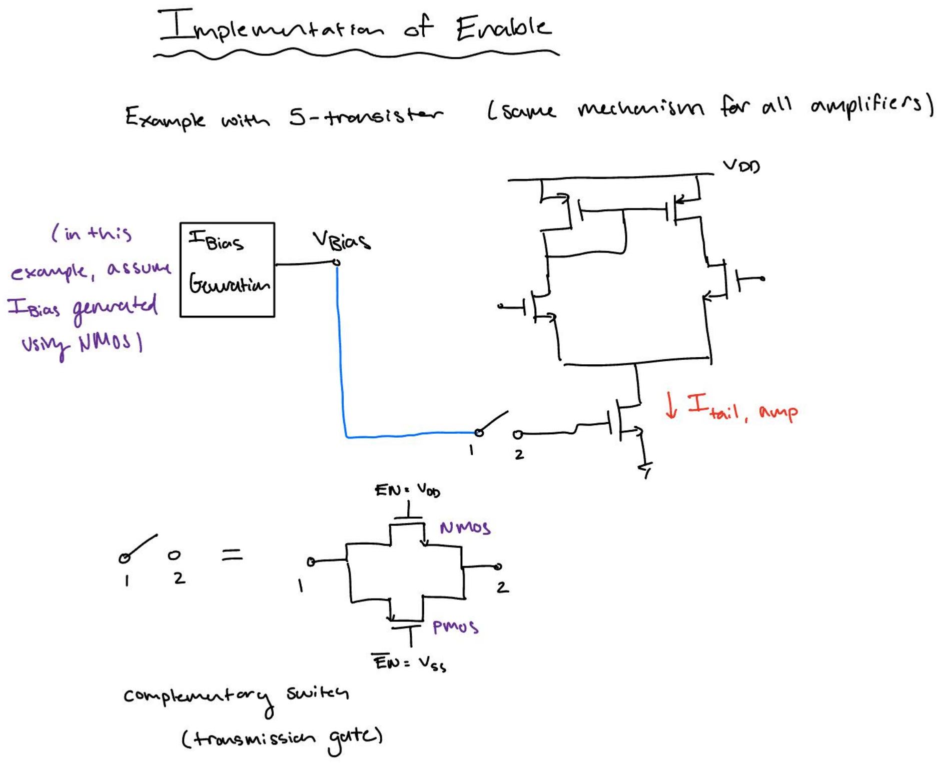
$$I_{Bias} \approx \frac{V_{DD} - |V_{GS}|_{M_1}}{R}$$

Project Information

Enable Implementation:

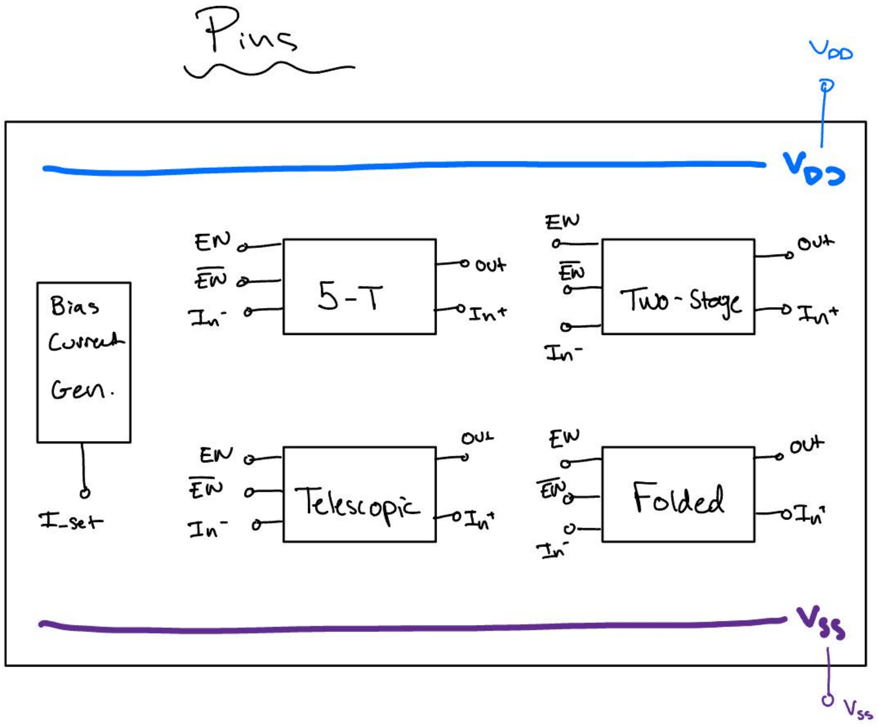
(GND should be VSS)

May rework, but principle



Project Information

Block Diagram:



(Will also need GND pin)

Team Background

Eli - **Coursework:** Advanced Analog Integrated Circuits, Communication Circuits, Analog Electronic Circuits, Principles of Device Microfabrication, Random Signals and Noise ; **Experience:** Analog Design Intern at Texas Instruments, ASIC Division Intern at BNL, Research Assistant in CLUE Lab, Teaching Assistant

Max - **Coursework:** Digital VLSI, Power Electronics, Semiconductor Device Physics, Principles of Device Microfabrication, Random Signals and Noise, Modern Display Science and Technology ; **Experience:** Hardware Developer Intern at IBM, High Voltage Electrical Engineering Intern, Research Assistant in the Bioelectronic Systems Lab, Teaching Assistant

Team Background

Manuel - **Coursework:** Communication Circuits, Analog Electronic Circuits, Random Signals and Noise, Modern Display Science and Technology ; **Experience:** Teaching Assistant, Independent Projects

Songhang - **Coursework:** Modern Display Science and Technology, Electronic Circuits ; **Experience:** Research Assistant in CLUE Lab, MIT Media Lab

Questions and Suggestions

- Is the bias current implementation reasonable, or should we add a current reference block?
- Is a more simple, fully analog MOSbius chip appropriate? It is currently more of a 'selectable' implementation, rather than a 'reconfigurable' one. Should we look to include some further digital control or programmability?
- What are some other things to keep in mind when designing, so that all amplifiers are 'optimized' about the same (so that one is not much better than the others, and it seems to dominate in all aspects and no meaningful comparisons can be made)? Currently, we are considering: holding the load constant for each amplifier, as well as matching total current
- Any other suggestions or feedback?

References

https://mosbius.org/0_front_matter/intro.html

Thank you track leads and mentors!